WORKING GROUP ON THE ASSESSMENT OF DEMERSAL STOCKS IN THE NORTH SEA AND SKAGERRAK (WGNSSK)

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International Council for the Exploration of the Sea
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H.C. Andersens Boulevard 44-46
DK-1553 Copenhagen V
Denmark
Telephone (+45) 33 38 67 00
Telefax (+45) 33 93 42 15
www.ices.dk
info@ices.dk

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Editors

Alessandro Orio • Lies Vansteenbrugge

Authors

Muhammad Abdullah • Elisa Barreto • Alessandra Bielli • Chun Chen • Harriet Cole
Mariana de Matos Domingues • José De Oliveira • Côme Denechaud • Raphaël Girardin
Ghassen Halouani • Holger Haslob • Rebecca Holt • Espen Johnsen • Graham Johnston • Claudia Jung
Alexander Kempf • Johan Lövgren • Tanja Miethe • Iago Mosqueira • J Rasmus Nielsen
Alessandro Orio • Yves Reecht • Jon Egil Skjæraasen • Klaas Sys • Marc Taylor • Justin Tiano
Lennert Van de Pol • Daniel van Denderen • Lies Vansteenbrugge • Damian Villagra Villanueva
Francesca Vitale • Nicola Walker

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i Executive summary

The main terms of reference for the ICES Working Group for the Assessment of Demersal Stocks in the North Sea and Skagerrak (WGNSSK) were to update, quality check, and report relevant data for the working group, to update and audit the assessment and forecasts of the stocks, to produce a first draft of the advice on the fish stocks, and to prepare planning for benchmarks in future years. Ecosystem changes have been analytically considered in the assessments for cod, haddock, and whiting in the form of varying natural mortalities estimated by the ICES Working Group on Multispecies Assessment Methods (WGSAM). Conservation status advice was considered but WGNSSK did not identify any conservation aspects for its stocks. From this year the *Nephrops* stocks have been removed from WGNSSK and will be assessed by a new working group WGNEPH.

Recent benchmarks

A benchmark that involved WGNSSK stocks was organized in 2024–2025. The WKBNSCS benchmark for North Sea turbot (tur.27.4) and for eastern English Channel plaice (ple.27.7d) evaluated the appropriateness of data and methods to determine stock status.

State of the stocks

The main impression in recent years is that fishing pressure has been reduced substantially for many North Sea stocks of roundfish and flatfish compared to the beginning of the century. Six stocks with agreed reference points (category 1 stocks) have SSB higher than MSY $B_{trigger}$ (haddock in 4, 6.a and 20, plaice in 4 and 20, sole in 4, turbot in 4, whiting in 4 and 7.d and witch in 3.a, 4 and 7.d). The SSB of saithe in 3.a, 4 and 6 is just above B_{lim}. Additionally, Northern Shelf cod (cod in 4, 6.a, 7.d and 20) has two substocks (Northwestern and Viking) with SSB below MSY $B_{trigger}$, and one substock (Southern) below B_{lim}. The SSB of Norway pout in 4 and 3.a, sole in 7.d and plaice in 7.d is below B_{lim}. Several North Sea stocks are exploited at or below F_{MSY} levels (haddock in 4, 6.a and 20, plaice in 4 and 20, sole in 4, sole in 7.d, turbot in 4, whiting in 4 and 7.d, and witch in 3.a, 4 and 7.d); however, plaice in 7.d, saithe in 3.a, 4 and 6, and all substocks of Northern Shelf cod (cod in 4, 6.a, 7.d and 20) are fished above F_{MSY}. An important feature is that recruitment remains poor compared to historical average levels for most gadoids, although there have been signs of strong recruitment for whiting in most recent years. However, recruitment in 2024 appears to be still low for the gadoids and lower than previous years for flatfish stocks, with the exception of plaice in 4 and 20, turbot in 4, and whiting in 4 and 7d.

WGNSSK is also responsible for the assessment of several data-limited stocks (category 2+ stocks) that are mainly bycatch in demersal fisheries; brill in 3.a, 4 and 7.d–e, lemon sole in 3.a, 4 and 7.d, dab in 3.a and 4, flounder in 3.a and 4, turbot in 3.a, whiting in 3.a, grey gurnard in 3.a, 4 and 7.d, striped red mullet in 3.a, 4, 7.d, pollack in 3.a and 4 and saithe in 7-10. Annual advice is required for brill in 3.a, 4 and 7.d–e, turbot in 3.a, and lemon sole in 3.a, 4 and 7.d. Biennial advice is required for striped red mullet in 3.a, 4 and 7.d, flounder in 3.a and 4, grey gurnard in 3.a, 4 and 7.d (due in 2026), and whiting in 3.a (due in 2026). Triennial advice is now required for dab in 3.a and 4, saithe in 7-10, and pollack in 3.a and 4 (due in 2027).

Category 2–6 fish stocks: In 2025, new advice has been produced for bll.27.3a47de and tur.27.3a (category 2), for dab.27.3a4, fle.27.3a4, lem.27.3a47d, mur.27.3a47d (category 3), and pok.27.7-10 (category 5). Advice was produced following the WKLIFE X rules. Advice was not provided for gug.27.3a47d, whg.27.3a (category 3), and pol.27.3a4 (category 5).

The summary of stock status is as follows:

Category 1 stocks

- Cod (cod.27.46a7d20): Spawning-stock biomass of the Northwestern substock has been above MSY $B_{trigger}$ in the most recent years and is just below in 2025. Spawning-stock biomass of the Viking substock has been fluctuating between MSY $B_{trigger}$ and B_{lim} in the most recent years and is below MSY $B_{trigger}$ in 2025. Spawning-stock biomass of the Southern substock has been below B_{lim} since 2017. Fishing pressure has been high throughout the time-series but decreased sharply from 2018 and is currently above F_{MSY} for all three substocks. Fishing pressure is also above F_{pa} for both the Viking and Southern substock. Recruitment (age 1) in recent years has been low for all three substocks, with 2025 being among the lowest recruitment in the time-series for all substocks.
- Haddock (had.27.46a20): Spawning-stock biomass has increased sharply since 2019 and is now well above MSY $B_{trigger}$. Fishing pressure has declined since the beginning of the 2000s, but it has been above F_{MSY} for most of the time-series. Since 2021, fishing pressure has been below F_{MSY}. Recruitment (age 0) since 2000 has been low with occasional larger year classes. Three recent year classes (2019, 2020, and 2022) have all been above the recent mean. Recruitment in 2024 is estimated to be low.
- Norway pout (nop.27.3a4): Spawning-stock biomass has been fluctuating over the time-series with a period of higher biomass in the 1990s, late 2000s, and around 2020. In the most recent year, SSB decreased, and is in 2025 below B_{pa} and B_{lim}. No reference points for fishing pressure or for MSY $B_{trigger}$ have been defined for this stock. Recruitment was below average in 2021 and is very low in 2023, 2024 and 2025..
- Plaice (ple.27.420): The spawning-stock biomass is well above MSY $B_{trigger}$ and has markedly increased since 2008, following a substantial reduction in fishing pressure since 1999. Currently, fishing pressure on the stock is below F_{MSY}. Recruitment (age 1) in 2022 is the second highest in the time-series, while recruitment in 2024 is the third highest.
- Plaice (ple.27.7d): The spawning-stock biomass has increased rapidly from 2010 following a period of higher recruitment between 2009 and 2019. However, since 2016, SSB has gradually declined and is now at a level just below B_{lim}. Higher fishing pressure from 2015 onwards and lower recruitment from 2021 onwards contributed to this decrease in SSB. Fishing pressure on the stock is currently situated above F_{MSY}. Recruitment (age 1) 2024 appears to be low.
- Saithe (pok.27.3a46): Spawning-stock biomass has mostly decreased since the early 2000s and is currently just above B_{lim}. Fishing pressure has decreased and stabilized above F_{MSY} since 2000. Currently, fishing pressure on the stock is above F_{MSY}. Recruitment (age 3) has shown an overall decreasing trend over time but slightly increasing in 2022. Recruitment in 2024 is low.
- Sole (sol.27.4): The spawning-stock biomass has fluctuated between MSY $B_{trigger}$ and B_{lim} since 2005, but is above MSY $B_{trigger}$ since 2021. Fishing pressure has been fluctuating above F_{MSY} for a large part of the time-series but sharply decreased from the beginning of the 2020s. In 2024, fishing pressure is below F_{MSY}. Recruitment (age 0) in 2018 was the 5th highest of the time-series. In more recent years, recruitment is estimated to be lower. Recruitment in 2024 is the highest since 2018.
- Sole (sol.27.7d): The spawning-stock biomass has decreased since 2014 and is estimated to be fluctuating around B_{lim} since 2017. Currently, SSB is estimated to be below B_{lim}. Fishing pressure has decreased since 2007, stabilised around F_{MSY} from 2019 onwards and is currently estimated below F_{MSY}. Recruitment (age 1) is estimated to be on a relatively low level since 2012 and the 2024 recruitment is slightly increasing compared to the preceding two years.

- Turbot (tur.27.4): The spawning-stock biomass has increased since 2004 and has been above MSY $B_{trigger}$ since 2007. Currently, spawning stock size is above MSY $B_{trigger}$. Fishing pressure has been above F_{MSY} from 2018-2021, but decreased after to levels below F_{MSY}. Recruitment (age 1) is variable without a trend. In 2024, it is estimated to be higher than the preceding 4 years.
- Whiting (whg.27.47d): Spawning-stock biomass has increased substantially in recent years and is now well above MSY $B_{trigger}$. Fishing pressure has been below F_{MSY} since the early 1990s. Currently, fishing pressure on the stock is well below F_{MSY}. Recruitment (age 0) has been fluctuating without trend, but the 2019 and 2020 year classes are estimated to be the largest since 2002. Recruitment in 2024 is estimated above average.
- Witch (wit.27.3a47d): Spawning-stock biomass has decreased and has been below MSY $B_{trigger}$ since the end of the 1990s but is above MSY $B_{trigger}$ in 2025. Fishing pressure has been above F_{MSY} since the beginning of the time-series. Currently, fishing pressure on the stock is at F_{MSY}. Recruitment (age 1) has decreased since 2019 and is at lower levels in 2024.

Category 2 stocks

- Brill (bll.27.3a47de): Catches declined sharply in recent years. The relative exploitable biomass (B/B_{MSY}) has been above the reference point since the beginning of the time-series. Relative fishing pressure (F/F_{MSY}) has decreased since 2000 and is currently below F_{MSY}.
- Turbot (tur.27.3a): Catches peaked in the late 1970s and early 1990s and have been more stable in recent years. The relative exploitable biomass (B/B_{MSY}) has been above reference point since the beginning of the time-series. Relative fishing pressure (F/F_{MSY}) peaked in the late 1970s and has been relatively low without a trend in more recent years. Currently, fishing pressure on the stock is below F_{MSY}.

Category 3 stocks

- Dab (dab.27.3a4): The stock size indicator has decreased since 2016 but is still above $I_{trigger}$. Currently, fishing pressure on the stock is below the $F_{MSY\ proxy}$.
- Flounder (fle.27.3a4): The stock size indicator has gradually decreased since 2008 and is below $I_{trigger}$ in 2025. Currently, fishing pressure on the stock is below the $F_{MSY\ proxy}$.
- Grey gurnard (gug.27.3a47d): rollover advice.
- Lemon sole (lem.27.3a47d): The stock size indicator has decreased since 2016 and is below $I_{trigger}$ in 2024. Currently, fishing pressure on the stock is below the $F_{MSY\ proxy}$.
- Striped red mullet (mur.27.3a47d): The stock size indicator has fluctuated above $I_{trigger}$ in the most recent years and is above $I_{trigger}$ in 2024. Currently, fishing pressure on the stock is above the $F_{MSY\ proxy}$.
- Whiting (whg.27.3a): rollover advice.

Category 5 stocks

- Saithe (pok.27.7-10): Catches have decreased steadily since 2013. Catches in 2024 are the lowest of the time-series.
- Pollack (pol.27.3a4): rollover advice.

Summary of retrospective analysis (WKFORBIAS decision tree)

To quantify retrospective patterns in the assessments of category 1 and 2 stocks, estimates of five-year retrospective peels are produced for fishing pressure, SSB and recruitment, and plotted with confidence bounds of the current assessment. The retrospective statistics (Mohn's rho) are reported as a measure of quality. The decision tree formulated by WKFORBIAS (ICES, 2020) was considered to ensure more consistency in how advice is provided.

ii Expert group information

Expert group name	Working Group on the Assessment of Demersal Stocks in the North Sea and Skagerrak (WGNSSK)
Expert group cycle	Annual
Year cycle started	2025
Reporting year in cycle	1/1
Chairs	Lies Vansteenbrugge, Belgium
	Alessandro Orio, Sweden
Meeting venues and dates	WGNSSK-1: 22 April to 2 May 2025, Copenhagen, DK (31 participants)
	WGNSSK-2: Autumn 2025, online meeting[1]

[1] Advice release scheduled for 10 October 2025 (Norway pout) and postponed stocks.

1 Introduction

1.1 Terms of reference

1.1.1 Generic ToRs for Regional and Species Working Groups

The generic ToRs are listed in Annex 2.

1.1.2 WG-specific ToRs

The specific ToRs for WGNSSK are listed in Annex 2.

1.2 Commercial data

1.2.1 Métier-based data call for WGNSSK

The year 2012 represented a major change in the process of data collection for WGNSSK. Following an initiative launched by ICES WGMIXFISH in August 2011, it had been decided to merge the data calls and data collection of both groups WGNSSK and WGMIXFISH, on the basis of:

1. Improving the availability of métier-based data and their consistency with the stock-based data used for single-stock assessment.
2. Allowing WGMIXFISH to meet earlier in order to integrate the mixed-fisheries advice within the single-stock advice sheets.

In 2014, data-limited stocks were included in the data call for the first time to improve the knowledge base for these stocks. With the landing obligation, these stocks become more important, and under these circumstances, discard information is a prerequisite for giving catch advice and carrying out mixed fisheries scenarios. In 2015, for the first time, a joint data call for all relevant assessment working groups was launched.

The principle of the data call is to define the aggregation (métier) level for the data that individual countries should deliver following the requirements of the EU Data Collection Framework (DCF) and to use these as the basis for providing and subsequently raising data for all North Sea demersal stocks. The ICES InterCatch database was chosen as the most appropriate tool to use until the planned Regional Data Base and Estimation System (RDBES) is fully established and operational. Basic strata for the submission of catch and effort data were by country, quarter, area, métier and catch category.

In 2019, the procedure for data submission was similar to previous years, including a requirement for life-history information and length compositions for historic landings and discards for stocks identified as “DLS” (essentially category 3 stocks) from at least the three most recent consecutive years (only the most recent year for those stock for which length frequency data were already provided in a previous data call). The data call also required reporting to four catch categories, including BMS landings (landings below minimum size for stocks under the landing obligation).

In 2020, in addition to the above procedure, coe.27.3a47de, hal.27.3a47de, and caa.27.3a47de were included in the data call to collect quarterly landings data for WGMIXFISH. The official data call was issued by ICES, with a deadline for data delivery of 1 April 2020, three weeks before the start of the WGNSSK meeting in Bergen. Despite delays in data submissions relative to the deadline and some errors needing to be corrected before the working group, these delays and corrections had no major impact on the work. During the meeting, it was noticed that 2019 landings for Sweden for Subarea 4 had not been uploaded to InterCatch. Amounts were generally low and were added manually for each affected stock to respective landings, and discards were raised using the discard ratio in area 4.

In 2021, the missing 2019 catches from Sweden were submitted, and catch data were re-raised and included in the respective assessments. Reductions in samples occurred for quarters 2–4 of 2020 and in quarters 1 and 2 of 2021 as a result of the COVID-19 pandemic. Any changes in the approaches for raising catch data in InterCatch related to reduced sampling were listed in the report (ICES, 2021a; ICES, 2022).

In 2023, the data call was adjusted following the WKBCOD benchmark in February 2023. Data on cod in the West of Scotland (cod.27.6a) was requested together with cod in the North Sea (cod.27.47d20). Moreover, ICES Member Countries were requested to supply 2022 commercial landings per quarter, cod stock area (substock) and ICES rectangle (Annex 7). In 2024, information on cod was requested as in 2023, with a later deadline compared to the other WGNSSK stocks. A specific template was provided to upload recreational catches for cod. Catch, sample and effort data were also requested for the new saithe stock (pok.27.7-10) under the WGNSSK data call, following the WKBGAD benchmark.

In 2025, no data was requested for *Nephrops* stocks in the WGNSSK data call, as they are now assessed in a separate WG. Prior to the WGNSSK data call deadline, the group was notified of a resubmission of data (landings, discards and distributions) by UK England for the years 2021-2023, covering all stocks on which UK England is fishing. While this resubmission had no major impact on the stock status of any WGNSSK stock, it did require all stock coordinators involved to repeat the raising and allocation processes, resulting in additional workload. Furthermore, the InterCatch platform, to which the data is submitted, experienced significant issues related to unit conversion errors. For example, landings were inadvertently multiplied by 1000 and mean weights at age were incorrectly copied into the mean length at age field. These errors required several countries to resubmit their data multiple times, and forced stock coordinators to delay or repeat the raising and allocation procedures.

1.2.2 Data raising and allocation to un-sampled strata

Major changes occurred over the last years with the raising of data within InterCatch. Different initiatives can be mentioned here:

1. Age and length data in parallel in InterCatch

InterCatch can work with age and length data in parallel, but it demands that length sample data have to be imported last for species with both age and length distribution data. This is due to InterCatch ignoring strata of other sample types. However, InterCatch will always take the latest imported strata without samples. Also, there is no problem with overwriting data in InterCatch as long as length data are imported latest, for stocks with both length and age samples. There is no option to upload age–length keys in InterCatch. It is important that when importing catches with and without age samples all strata have to be imported, all strata also have to be imported when importing catches with and without length samples.

2. Technical improvements in the InterCatch interface

- Allocation Group Setup: define a group of unsampled catch/strata for which each distribution will be calculated according to the (for the group) allocated sampled catches/strata;
- Automatic allocation of 'same' strata: automatically find and allocate identically sampled strata from other countries to unsampled catches/strata (with the identical stratum);
- Discard Group setup: Define a group of raised discards for which each discard weight will be calculated according to the (for the group) selected landing-discard ratios;
- CATON and age/length data overviews: it is possible to examine all imported data in detail;
- Allocation overview for pivot table/matrix: all unsampled strata are shown in the first column and all sampled strata are shown as the first row, then all the selected combinations are shown in the matrix;
- Possibility to save allocation schemes.

3. Summary outputs and inspection of data before raising

The features included in InterCatch allow inspection and visualization of the data submitted by national data providers and a comparison with data from previous years. A generic R script was developed in 2016 and improved in subsequent years by Y. Vermard (Ifremer) mapping out the raw data, through e.g. quantification of the proportion of catches covered by sampling, identification of major gaps and outliers, plot of the age distribution and discards ratio of the various strata etc.

4. Raising procedures

Based on statistical principles discussed within WKPICS, RCMs, PGCCDBS, DC-MAP etc., the suggestions for the basis on which to proceed regarding raising of age distributions and discards ratio have been revisited. In 2012, the raising and allocating was based on finding similar strata from other countries, but this was judged not fully defendable in terms of statistical integrity. In 2016, the underlying principles applied were thus:

- Main strata are supposed to be sampled. In essence, one should expect that the largest share of catches should have age-based and discards information in InterCatch. Although there may be a great number of unsampled strata, in reality, these should represent only a minor part of the catches. Large strata without sampling information would need to be investigated further.
- Therefore, the suggestion was that by default, unsampled strata should be raised by all sampled strata, unless there is a good and informed reason for choosing differently after the data inspection process. Each stock coordinator has developed general principles for the allocation scheme. The main principles are mentioned in the respective report sections.

Ultimately, all these changes have triggered an in-depth investigation and understanding of the data submitted, and it is hoped to contribute to improved consistency and transparency in the assessment data. However, if more than one year needs to be raised, the InterCatch procedure is very time-consuming. The saving of allocations schemes does not always function, especially when the métiers differ between years, and currently, only the age allocation scheme can be copied (not the discard ratio allocation scheme). It would be beneficial to allow for more flexible automatic matching based on e.g. gear type or area only. Also, the possibility of entering allocation schemes via scripts (instead of the need to click through the options and métiers) would allow for fast sensitivity checks and would make InterCatch much more user-friendly. However, there is limited scope for improvements in InterCatch, given the focus on getting RDBES (its successor) operational and fully functional in the near future. Several workshops have been organized or are planned to facilitate the switch from InterCatch to RDBES. WKRDBES-

RaiseTAFFlow (22–26 May 2023) focused on establishing the data flow and access roles from national submissions over discard raising and age allocation to running the assessment of two WGNSSK stocks: saithe (pok.27.3a46) and witch flounder (wit.27.3a47d). Several follow-up workshops have been planned in 2024, with the WGRDBES-StockCoord planned in October 2024 particularly interesting to stock coordinators to further develop R scripts/functions for stock estimation. Some of the WGNSSK stock coordinators already fully switched from using InterCatch for raising and allocation procedures to R scripts, an example is North Sea sole following the recent WKBFLATFISH benchmark (February 2024; ICES, 2024). In 2025, this work continues under the umbrella of WGRDBES-StockCoord, in the form of quarterly meetings.

Because of the landing obligation, new catch categories have been reported since 2016. BMS landings, observer discards and logbook recorded discards should sum up to discard data provided before 2016 (i.e. double-counting should be avoided), and when performing raising procedures, the raising procedure in InterCatch should be adapted as necessary to provide a robust approach, independent of how countries categorize catches when providing catch data. The general approach adopted by WGNSSK is to raise discards using only the observed discards (catch category "D" from the data call), and to allocate discard age compositions to BMS landings (category "B" from the data call) if reported and given a "CATON" value.

Similarly, no discard estimates are calculated for the industrial bycatch (IBC) strata and age/length allocation for the IBC strata is done using all available information from both landings and discards.

InterCatch summary data have been made available on SharePoint and will be investigated further during ICES WGMIXFISH.

In October 2025, the status of InterCatch use was as follows:

Stock	Data Year	Working Group	Extracted	Exported	Status of Data filled in
bll.27.3a47de	2024	WGNSSK	Extracted	Exported	DataUsedForAssessment
cod.27.46a7d20	2024	WGNSSK	No	No	Notfilled
cod.27.47d20	2024	WGNSSK	Extracted	Exported	DataUsedForAssessment
dab.27.3a4	2024	WGNSSK	Extracted	Exported	DataUsedForAssessment
fle.27.3a4	2024	WGNSSK	Extracted	Exported	DataUsedForAssessment
gug.27.3a47d	2024	WGNSSK	Extracted	Exported	DataUsedForAssessment
had.27.46a20	2024	WGNSSK	Extracted	Exported	DataUsedForAssessment
lem.27.3a47d	2024	WGNSSK	Extracted	Exported	Notfilled
mur.27.3a47d	2024	WGNSSK	Extracted	Exported	DataNOTusedForAssessment
nop.27.3a4	2024	WGNSSK	No	No	Notfilled
ple.27.420	2024	WGNSSK	Extracted	Exported	DataUsedForAssessment
ple.27.7d	2024	WGNSSK	Extracted	Exported	DataUsedForAssessment
pok.27.3a46	2024	WGNSSK	Extracted	Exported	DataUsedForAssessment
pok.27.7-10	2024	WGNSSK	Extracted	No	Notfilled

Stock	Data Year	Working Group	Extracted	Exported	Status of Data filled in
pol.27.3a4	2024	WGNSSK	Extracted	Exported	DataUsedForAssessment
sol.27.4	2024	WGNSSK	Extracted	No	Notfilled
sol.27.7d	2024	WGNSSK	Extracted	Exported	DataUsedForAssessment
tur.27.3a	2024	WGNSSK	Extracted	No	Notfilled
tur.27.4	2024	WGNSSK	Extracted	Exported	Notfilled
whg.27.3a	2024	WGNSSK	Extracted	No	Notfilled
whg.27.47d	2024	WGNSSK	Extracted	Exported	DataUsedForAssessment
wit.27.3a47d	2024	WGNSSK	Extracted	Exported	DataUsedForAssessment

1.2.3 Issues with landings data

For a number of years, there have been indications that substantial misreporting of roundfish and flatfish landings is likely to have occurred. It is suspected to have been particularly strong for cod until 2006, and catches were expected to be larger than the TAC. Since the middle of the 2000s, the WG had used an assessment method for North Sea cod (Section 4) which estimated unallocated removals, potentially due to reporting problems, unrecorded discards, changes in natural mortality, or changes in survey catchability. In 2013, WGNSSK considered that the assumption of unallocated removals after 2006 could not be justified by any known factors (also see ICES WKCOD, 2011), and relaxed that assumption (from 2006 onwards) in the assessment.

Similarly, landings of the Belgian fleet are corrected for sole in the eastern English Channel (sol.27.7.d) for the period 2004–present since the benchmark in 2021 (ICES, 2021b).

Commercial CPUE or LPUE series were available for a number of fleets and stocks, but for various reasons, only some of them could be used for assessment purposes (although they are presented and discussed). The use of these commercial indices has been phased out where possible and of the eleven category 1 assessments, only saithe in 3a46 and sole in 7.d include one and three commercial indices respectively.

1.2.4 Issues with discards and BMS landings

Estimates of discarding rates provided by a number of countries through observer sampling programmes were used in the assessments of various roundfish and flatfish, to raise landings to catch (see also §1.2.2). Discards could also be estimated for bycatch species (e.g. dab, flounder, grey gurnard, lemon sole, witch, saithe in 7-10, brill, and turbot). Finally, catch advice could be given for all WGNSSK stocks that require it.

Despite the EU landing obligation and Norway and UK national legislation regulating discards, landings of fish below the minimum size (BMS) reported to ICES are very low and discarding still takes place. There remain inconsistencies in the reporting of BMS landings between different nations, both in the official statistics (FAO) and in InterCatch. In general, WGNSSK has assumed that BMS landings are part of discards, and BMS landings are not shown separately in tables of ICES estimates given in the advice sheets; the only BMS estimates that appear in advice sheet tables are those from official statistics. The only exceptions to this treatment of BMS landings as discards are for the saithe stock (pok.27.3a46), for which the Norwegian component of BMS landings is included with the ICES estimates of landings issues with missing hauls in the central/northern North Sea and West of Scotland.

1.2.5 Issues with survey data

Similar to 2023, surveys didn't encounter many coverage issues in 2024. A small number of planned hauls was not sampled in 2024, due to algal blooms, bad weather or technical issues, but the impact on the assessments was negligible. The French YFS 2024 data could not be provided for the sole in 7d assessment, however, impact was also negligible. Some surveys were carried out by different vessels, but this was accounted for in the survey models.

Surveys encounter more and more restrictions related to marine protected areas (MPAs) and offshore renewable energy (OREs). Several workshops will be organised in 2025 to analyse the impact of and determine alternatives to deal with these spatial changes (e.g. WKDISM June 2025, WKUSER3 October 2025).

WGNSSK, as one of the end-users, finds the recent development of a DATRAS Data submission tool a big step forward in keeping track of the changes made in the DATRAS database. The tool was demonstrated again during the WGNSSK spring meeting.

1.3 General uncertainty considerations

Data or inputs used in this report are based on sampling or census. Typical census data are landings data from sales slips representing total landings, while sampled data are random samples (design-based) used to produce estimates of total, relative indices or to characterize composition (like catch-at-age). All sources of input may introduce errors in estimates/calculations and are a limiting factor in the amount of signal in data and/or interpretation of model results. The scientists at this working group are only responsible for a modest fraction of the input data used and are relying heavily on assumptions regarding their validity and quality. The information based on sampling will contain sampling errors (random errors due to the stochastic nature of such sampling) and estimates of sampling error are generally not used by this working group. Such errors will show up in residuals (residual plots are an important diagnostic in the report), but other sources of error will also show up in the same residuals and are not easily separated from random errors. Non-random errors are either bias or model errors. Systematic bias over time is a particular concern and an example of such can be underreporting of catches, which will compromise the validity of the model results as basis for advice. Model errors may represent the use of the "wrong" equations to describe relations, but will in this report typically be linked to assumptions regarding natural mortality, the relationship between survey indices and stock size (catchability) and exploitation pattern. Some assumptions are needed since, for example, the Baranov catch equations do not have unique solutions (too many parameters to estimate).

Assessment working groups are in many ways end-users of data and it would be preferable to have such information presented as point estimates together with estimates of uncertainty or confidence bands and with a description of potential sources of bias and qualitative remarks related to specific observations. Where InterCatch does not fully meet these requirements, the RDBES might.

The working group appreciates the effort made by so many supporting hands involved in creating all information needed in fish stock assessment and depends on the quality of information being upheld over time. An assessment working group is where information from the commercial fishery is handled together with fishery-independent information to create estimates of stock status and the impact of fishing.

Demersal trawl surveys are the most used source of fishery-independent information in this working group (WGNSSK). A demersal trawl survey uses a standardized procedure of trawling to create samples from a fish population. The “population” in statistical terms is the population of possible trawl stations with the trawl station being the primary sampling unit. The estimates of uncertainty from a demersal trawl survey are very much dependent on the number of samples (trawl stations) and it seems that demersal trawl surveys on gadoids produce very similar estimates of uncertainty given the same number of trawl stations (ICES, 1992) regardless of the size of the area. The relationship between sample size and precision can be illustrated using the following example: If a survey of 400 trawl stations produces an estimate (for a parameter of interest) with a corresponding relative standard error of 0.1 a reduction in survey effort to 100 trawl stations is likely to produce estimates with a relative standard error of 0.2 (divide the number of stations by 4 and the relative standard error is doubled). This is also likely to hold (at least as a rule of thumb) if one looks at results from a subarea of the original (400 stations) area. When estimates of relative standard error approach 0.3, trends over time will be very difficult to detect, and with relative standard errors above 0.3, the estimator can only be used to detect sudden events. WGNSSK recommends that, along with survey index point estimates, DATRAS should also provide the uncertainty around these estimates as standard output.

1.4 Internal auditing and external reviews

Although a very important quality assurance mechanism, internal audits do place an additional burden on group members, and it has not been possible to complete audits during the meeting itself for a few years now. WGNSSK operates with seldom more than one scientist per stock (sometimes one scientist is responsible for two or more stocks), and there was in most cases not enough time to have the reports finalized in order to carry out the audit within the WG meeting itself. Audits had to be conducted by correspondence after the WG time, which is neither very efficient nor very motivating, given the heavy workload under which most members usually operate back at home institutes. It is hoped that the move to TAF will both make auditing easier and more transparent and improve the quality of auditing procedures.

In addition to the internal audits, WGNSSK introduced an additional quality check this year to further strengthen the quality of the advice for all category 1 stocks. This quality check is part of the checks WGMIXFISH carries out in autumn to generate their advice. All stocks involved passed this quality control.

All WGNSSK stocks with advice in 2025 could be covered by the internal audit (Table 1.4.1). The audit results are provided in Annex 4 of the report.

External reviews were required for had.27.46a20, ple.27.7d, and tur.27.3a in 2025 and can be found in Annex 8 of the report (Working Documents 1, 7, and 8).

Table 1.4.1. Fish stocks covered by the internal audit.

Fish Stock	Internal Audit Spring	Internal Audit Autumn
bll.27.3a47de	X	
cod.27.47d20	*No advice in spring*	
dab.27.3a4	X	
fle.27.3a4	X	
gug.27.3a47d	*No new advice in 2025*	

Fish Stock	Internal Audit Spring	Internal Audit Autumn
had.27.46a20	X	
lem.27.3a47d	X	
mur.27.3a47d	X	
nop.27.3a4	*No advice in spring*	X
ple.27.420	X	
ple.27.7d	X	
pok.27.3a46	X	
Pok.27.7-10	X	
pol.27.3a4	*No new advice in 2025*	
sol.27.4	*No advice in spring*	X
sol.27.7d	X	
tur.27.3a	*No advice in spring*	X
tur.27.4	X	
whg.27.3a	*No new advice in 2025*	
whg.27.47d	X	
wit.27.3a47d	X	

1.5 Transparent Assessment Framework (TAF)

TAF is a framework to organize all ICES stock assessments. Using a standard sequence of R scripts, it makes the data, analysis, and results available online, and documents how the data were preprocessed. Among the key benefits of this structured and open approach are improved quality assurance and peer review of ICES stock assessments. Furthermore, a fully scripted TAF assessment is easy to update and rerun later, with a new year of data. As of spring 2018, the first assessments have been scripted in standard TAF scripts[2]. By 2020, 14 out of 30 WGNSSK stocks were in varying states of completeness in TAF. Uptake has then decreased. Stock assessors were encouraged to take part in training workshops offered by ICES to get their assessments into TAF. At the end of WGNSSK 2025, 15 out of 21 stocks are in TAF and more are planning to work towards the TAF implementation in the near future.

1.6 Special request and technical service

WGNSSK was involved in the special requests on Northern Shelf cod:

[2] For more information, see: http://taf.ices.dk

The **Workshop on Scoping Data collection for Northern Shelf cod sub-stocks (WKCODSCOPE)** is held in response to a special request from the United Kingdom, Norway and the European Commission (EC, DGMARE).

WKCODSCOPE addressed several topics to scope the Northern shelf cod sub-stock data needs:

- Summarize the current sub-stock definition for the Northern Shelf cod complex and review the progress of the project GenDC,
- Outline additional genetic data needs to further understand the sub-stock mixing throughout the year to facilitate the provision of advice which can guide area-specific management,
- Develop sampling strategies according to a partial or complete separation of northwestern and southern components of the Northern Shelf cod complex.
- Explore alternative stock discrimination techniques (otolith shape and microchemistry, tagging, drift-based, morphological or phenotypical data) and their potential to elucidate sub-stock spatial dynamics throughout the year.

The workshop took place in March 2025. A continuation of this workshop will be organised at the end of 2025 - beginning of 2026.

1.7 Presentations

One presentation was given to WGNSSK in 2025. Below is the summary of the presentation.

1.7.1 Stability and drivers of Atlantic cod subpopulations in the North Sea and adjacent

Hsiao-Hang Tao[1]*, Manuel Hidalgo[2], Vasilis Dakos[1]

[1] Institut des Sciences de l'Evolution de Montpellier, Université de Montpellier, Montpellier, France

[2] Spanish Institute of Oceanography (IEO, CSIC), Balearic Oceanographic Center, Group of Ecosystem Oceanography (GRECO), Moll de Ponent s/n, 07190, Palma, Spain

*corresponding: hsiaohang.tao@gmail.com

The dynamics and resilience of Atlantic cod (*Gadus morhua*) in the North Sea and adjacent waters have been extensively studied due to the population's ecologic and economic importance, as well as its marked decline since 2000. Recent evidence indicates that this population consists of three distinct subpopulations—northwestern, southern, and viking— with limited mixing between them. Since 2024, stock assessments have been conducted separately for each subpopulation, motivating further research into the dynamics and resilience at the subpopulation level to support more targeted fisheries management.

In this study, we applied empirical dynamic modeling (EDM), which accounts for state-dependent and nonlinear population dynamics, using yearly abundance time series data for seven age groups from 1983 to 2023. We quantified two key properties for each subpopulation: stability, reflecting the tendency to stabilise at or diverge from the abundance of each year, and relative sensitivity of age groups to perturbations, indicating the relative change in abundance of each age group in response to a perturbation or fluctuation of all age groups. We then examined the effects of fishing and environmental change on the abundance and stability of each subpopulation.

Our results show that the temporal changes in stability differed across the three subpopulations. The northwestern subpopulation switched from a disequilibrium state to a stabilised state at a low abundance. The southern subpopulation shifted from a relatively high-abundance stable state towards critical transition when the abundance declined abruptly. This suggests an ongoing change that could result in further decrease in abundance or instead an increase in abundance. In contrast, the viking subpopulation remained stable with little change in abundance over time.

In terms of the relative sensitivity between age groups, the age groups in the northwestern subpopulation displayed similar sensitivity over the study period, whereas age-1 group was frequently the most sensitive in both the southern and viking subpopulations. Fishing mortality was found to causally affect the abundance of the northwestern subpopulation and to influence both the abundance and stability of the southern subpopulation. However, it showed no causal relationship with the viking subpopulation. Sea bottom temperature did not show any causal influence on abundance or stability across any of the subpopulations.

Overall, our findings reveal distinct patterns of abundance, stability, sensitivity, and regulatory mechanisms among the three cod subpopulations, emphasizing the importance of accounting for the complex structure of this population in future studies to support effective management strategies.

1.8 References

ICES. 1992. Report of the workshop on the analysis of trawl survey data. Woods Hole, 4.-9. June 1992. C.M. 1992/D:6.

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ICES. 2022. Working Group on the Assessment of Demersal Stocks in the North Sea and Skagerrak (WGNSSK). ICES Scientific Reports. 4:43. 1367 pp. http://doi.org/10.17895/ices.pub.19786285

ICES. 2024. Benchmark workshop on selected flatfish stocks (WKBFLATFISH). ICES Scientific Reports. 6:30. 729 pp. https://doi.org/10.17895/ices.pub.25471987

2 Overviews

2.1 Fisheries overview

2.1.1 Summary

The demersal fisheries in the Greater North Sea ecoregion can be categorized as a) human consumption fisheries, and b) industrial fisheries which land the majority of their catch for reduction purposes. Demersal human consumption fisheries usually either target a mixture of roundfish species (cod, haddock, whiting) or a mixture of flatfish species (plaice and sole) with a bycatch of roundfish and other flatfish (e.g. turbot, brill, dab). A fishery directed at saithe with some bycatch of hake and other roundfish exists along the shelf edge. Main North Sea stocks targeted in the fisheries for industrial purposes are sandeel, Norway pout, and sprat.

The total demersal landings from the Greater North Sea peaked above 1.5 million tonnes in the period 1970–1990, showed a strong decline from the mid to late 1990s, and fluctuated now around 500 000 tonnes (ICES, 2024a). The industrial fisheries which used to dominate the North Sea demersal catch in weight have become much less prominent. Landings of the industrial fisheries show the largest annual variations, resulting from variable recruitment and the short lifespan of the main target species. Human consumption landings have steadily declined over the last 40 years.

Gear types vary between fisheries (ICES, 2024a). Human consumption fisheries use otter trawls, pair trawls, seines, gillnets, or beam trawls, while industrial fisheries use small-meshed otter trawls which in most cases are equipped with selective panels to reduce bycatches.

For some stocks, the assessment area covers other regions adjacent to the Greater North Sea ecoregion. Thus, combined category 1 assessments are made for haddock including Division 6.a, saithe including Subarea 6 and cod including Division 6.a.

2.1.2 Mixed fisheries

The mixed fisheries analyses for the North Sea are performed by the Working Group for Mixed Fisheries Advice (WGMIXFISH-ADVICE; ICES, 2025), which aims to evaluate the consistency of the ICES advice for the individual stocks in a mixed fisheries context, using a FLBEIA model (Garcia *et al.*, 2017), as part of the FLR framework (Kell *et al.*, 2007; www.flr-project.org). WGMIXFISH meets directly after WGNSSK in June 2025 (WGMIXFISH-METH), and also in October 2025 (WGMIXFISH-ADVICE). We, therefore, refer to the ICES WGMIXFISH reports and Fisheries Overview for any further description of the mixed-fisheries context.

WGNSSK and WGMIXFISH have developed and issued a common data call since 2012, which has greatly improved the quality and scheduling of data delivery (Annex 7). The data distinguish between two basic concepts, the Fleet (or fleet segment), and the Métier. Their definition has evolved with time, but the most recent official definitions are those from the European Commission's Data Collection Framework (DCF, Reg. (EC) No 949/2008), which we adopt here*:*

- A **Fleet** segment is a group of vessels with the same length class and predominant fishing gear during the year. Vessels may have different fishing activities during the reference period but might be classified in only one fleet segment.

- A **Métier** is a group of fishing operations targeting a similar (assemblage of) species, using similar gear, during the same period of the year and/or within the same area and which are characterized by a similar exploitation pattern.

Fleets and métiers were defined to match the available economic data and the former cod long-term management plan.

In 2024, mixed fisheries advice considerations included brill (bll.27.3a47d), cod (cod.27.46a7d20), haddock (had.27.46a20), plaice (ple.27.420 and ple.27.7d), saithe (pok.27.3a46), sole (sol.27.7d), turbot (tur.27.4), whiting (whg.27.47d), witch (wit.27.3a47d), and Norway lobster (FUs 5–10, 32, 33, 34, and Subarea 4 outside the FUs). According to the WGMIXFISH 2024 results, based on current fishing patterns and single-stock catch advice, the most limiting stock for North Sea demersal fisheries is cod, whose advised catch for 2025 is first reached for 23 of 37 defined fleets. Whiting is the least limiting stock in 35 of 37 fleets making it overall the least limiting stock. The advised catches for these stocks imply very different changes in effort level in 2025 relative to *status quo*. The largest reduction in effort across most fleets relative to *status quo* is seen for cod, which is the most limiting stock for North Sea demersal fisheries (the stock where catch advice implies the lowest level of effort by the fleets). Other stocks which show a reduction in effort across most fleets are plaice in the eastern English Channel, sole in the eastern English Channel, witch, and Norway lobster in FUs 6, 32, 33, 34, and Subarea 4 outside the FUs. Conversely, an increase in effort is seen across most fleets for brill, haddock, North Sea plaice, saithe, turbot whiting and Norway lobster in FUs 5, and 7–10.

ICES has evaluated technical interactions between species captured together in demersal fisheries by examining their co-occurrence in the landings at the scale of gear/mesh size range/ICES square/calendar quarter (hereafter referred to as ‘strata’). The percentage of landings of species A, where species B is also landed and constitutes more than 5% of the total landings in that stratum, has been computed for each pair of species. Cases in which species B accounts for less than 5% of the total landings in a stratum were ignored.

To illustrate the extent of the technical interactions between pairs of species, a qualitative scale was applied to each interaction (Figure 2.1.1). In this figure, rows represent the share of each species A that was caught in fisheries where species B accounted for at least 5% of the total landing of the fisheries. For example, a large proportion of the catches of lemon sole was taken in fisheries where plaice landings constituted at least 5% of the total landings; medium quantities of lemon sole were caught in fisheries where cod, haddock, hake, or saithe accounted for at least 5% of the total landings; low quantities of lemon sole were caught in fisheries where lemon sole constituted 5% or more of the total landings, indicating that there is no (or very limited) targeted lemon sole fishery.

The columns in Figure 2.1.1 illustrate the degree of mixing and can be used to identify the main fisheries (target fisheries) and the degree of mixing within these fisheries. Fisheries where plaice (species B) constitute 5% or more of the total landings account for a high share (red cells) of the total landings of dab, lemon sole, plaice, sole, turbot, flounder, brill, haddock, and which, and a medium share (orange cells) of the landings of whiting, hake and *Nephrops*. This shows that the plaice fishery is a central fishery in the North Sea with a high degree of mixing. The lemon sole column shows that the landings of lemon sole, in fisheries where the species constituted 5% or more of the total landing, were low, and the relative landings of other species in these fisheries were also low. This confirms that there is no or very limited targeted lemon sole fishery.

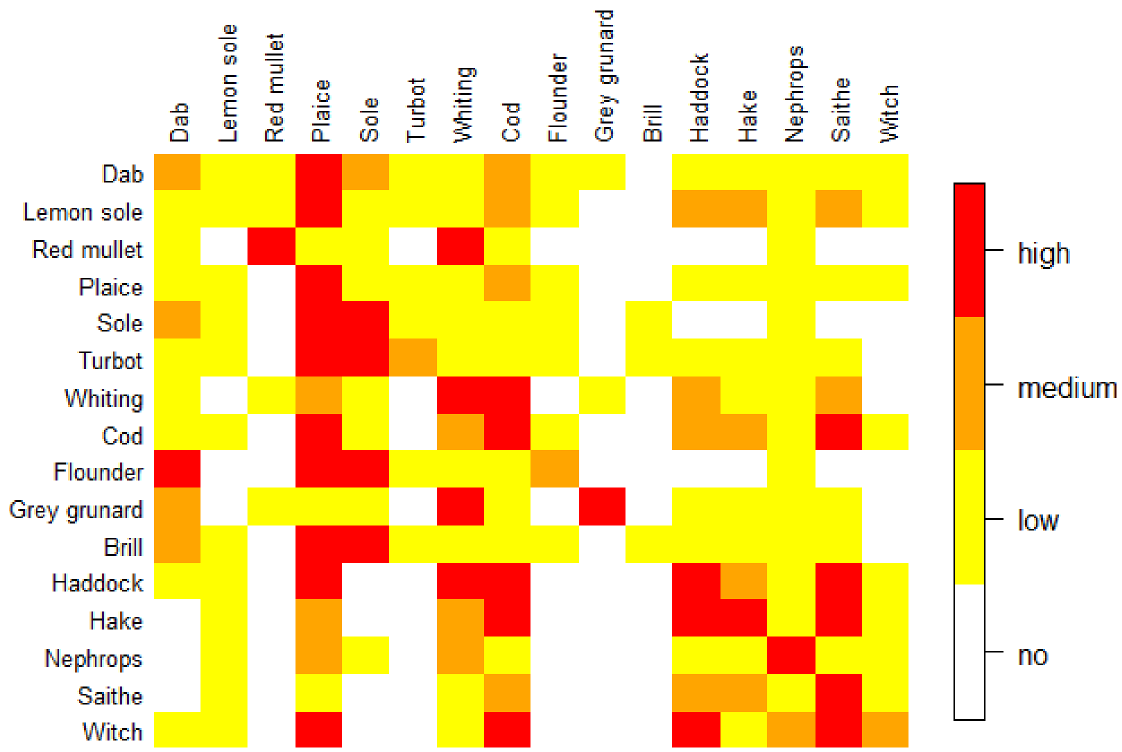

Figure 2.1.1. Technical interactions among Greater North Sea demersal stocks. The rows illustrate the fisheries where species A was caught. Red cells indicate species B (listed as the columns) with which species A are frequently caught. Orange cells indicate medium interactions and yellow cells indicate weak interactions. The column shows the degree of mixing in fisheries where species B accounts for at least 5% of the total landings. A more detailed explanation of the figure is provided in the text.

2.1.3 Audit of Fisheries Overview (ToR i)

ICES published a Fisheries Overview for the Greater North Sea Ecoregion in December 2025 (ICES, 2024a). No new audit was done in 2025.

2.2 Main management regulations

The near collapse of the North Sea cod stock at the beginning of the 2000s led to the introduction of effort restrictions alongside TACs as a management measure within EU fisheries. There has also been an increasing use of single-species multiannual management plans, partly in relation to cod recovery, but also more generally. With the implementation of the landing obligation in 2016, EU multiannual plans were developed and are available for North Sea demersal stocks (Regulation (EU) 2018/973) and stocks fished in western waters (Regulation (EU) 2019/472). Further, technical measures regulations have been amended (Regulation (EU) 2019/1241), e.g. the prohibition of electric pulse trawl since 1 July 2021. Finally, from 2024 onwards, the TAC for North Sea turbot and brill, and lemon sole and witch flounder no longer cover multiple species and was aligned to comprise the areas covering the whole stock (EU, 2023).

The management frameworks can be summarized as such:

2.2.1 Landing obligation

Fisheries in Norwegian waters have been subject to a landing obligation for cod and haddock since 1987 and for most species since 2009. A landing obligation for EU fisheries on demersal species in the North Sea was implemented in 2016 in a phased approach with all quota stocks subject to the landing obligation from 2019 onwards. Detailed definitions of the landing obligation can be found in Article 15 of Regulation 1380/2013. Discard plans have been agreed for 2018 in the North Sea (Subarea 4, Division 3.a and Union waters of Division 2.a; Regulation (EU) 2018/45) and in Union and international waters of Subarea 6 and Division 5.b (Regulation (EU) 2018/46), and in Division 7.d (Regulation (EU) 2018/46), defining for which species, gear and mesh size combinations the landing obligation applies. These have been updated for 2019–2021 (Regulation (EU) 2018/2035 and Regulation (EU) 2018/34) to reflect that all demersal quota stocks are now subject to landings obligations, but also to detail survivability and *de minimis* exemptions and specific technical measures. In 2019, new updates were published for 2020–2021 (Regulation (EU) 2019/2238 and Regulation (EU) 2019/2239), to modify in part the details of survivability and *de minimis* exemptions and specific technical measures. In 2023, new updates for the period 2024–2027 were published (Commission Delegated Regulation (EU) 2023/2459).

There is a high probability that the implementation of the EU landing obligation with its complex definitions, exemptions and rules (e.g. *de minimis*, high survival, 9% interspecies flexibility) has implications for the quality of monitoring of the catches and the quality of assessments of the stock status and exploitation rate. *De minimis* exemptions and the 9% interspecies flexibility rule may have serious implications for stocks dependent on the interpretation of the respective paragraphs in the regulation (STECF, 2014a; STECF, 2014b).

The data provided to ICES does not include information that would allow ICES to evaluate the impact or take account of the complex survivability and *de minimis* exemptions. Furthermore, there was no evidence presented to the Working Group that the introduction of the landing obligation had caused any change to discarding practices for fisheries since 2016.

For sole, plaice, turbot and haddock, several *de minimis* or high survival exemptions have been agreed. The default ICES assumption is that the same exploitation patterns will continue. How much of the unwanted catch is landed (catch category BMS) and how much is still discarded is speculation. Given that stocks are impacted by the total F independent of how the total catch is split up (at least under the assumption of no survival of discards), the results of forecasts are robust to assumptions regarding which fraction of the total catch will be landed. It is unclear how exploitation patterns of fleets have changed so far. This leads to an increased uncertainty in short-term forecasts. ICES has the incentive to further investigate the assumption of discard survivability and explore accounting for this in stock assessments (survivability roadmap).

It would be expected that under the EU Landing Obligation, fish caught under the minimum conservation reference size (MCRS) would be landed and recorded as BMS landings in logbooks rather than discarded as happened before the Landing Obligation. The logbook records of BMS landings would then be reported to ICES. However, low BMS values may be seen if the fish caught below MCRS are either not landed, not recorded in logbooks, not reported to ICES, reported to ICES incorrectly, or a mixture of any of these. For all stocks where BMS landings were reported to ICES since 2016, these values were either zero or very low, substantially lower than the estimated discards.

2.2.2 Effort limitations

For vessels registered in EU member states, effort restrictions in terms of days at sea were introduced in 2003 and subsequently revised annually. Initially, days at sea allowances were defined by calendar month. From 2006, the limit was defined on an annual basis. The maximum number of days a fishing vessel could be absent from port varied according to gear type, mesh size (where applicable) and region. A complex system of 'special conditions' (SPECONs) developed upon request from the Member States, whereby vessels could qualify for extra days at sea if special conditions (specified in the Annexes) were met. Increasingly detailed micromanagement took place until 2008 (Ulrich *et al.*, 2012).

In 2008, the system was radically redesigned. From 2009, a total effort limit (measured in kW days) was set and divided up between the various nation's fleet effort categories. The baselines assigned in 2009 were based on track record per fleet effort category averaged over 2004–2006 or 2005–2007 depending on national preference, and the effort ceilings were updated in 2010. After some reductions based on the cod management plan to support the recovery of the cod stock, an effort rollover for the maximum allowable fishing effort was decided for 2013–2016. The effort management regime, which formed part of the long-term management plan for North Sea cod, has been revoked from 2017 onwards. The effort management regime for plaice and sole continued to apply in 2018 while the second stage of the management plan (Council Regulation (EC) 676/2007) was still in place. The effort management regime for plaice and sole has been revoked from 2019 onwards with the implementation of the EU MAP for sole (Regulation (EU) 2018/973).

The STECF and ICES WGMIXFISH have performed annual monitoring of deployed effort trends since 2002. In addition, a more detailed overview and analyses of the various measures implemented in the frame of the cod recovery plan can be found in the 2011 joint STECF/ICES evaluation of this plan (ICES WKROUNDMP, 2011; Kraak *et al.*, 2013).

2.2.3 Stock-based management plans

Cod, haddock, whiting, saithe, plaice, and sole have previously been subject to multiannual management strategies (the latter two, being EU strategies, not EU-Norway-UK agreements). These plans all consist of harvest rules to derive annual TACs depending on the state of the stock relative to biomass reference points and target fishing mortalities. The harvest rules also impose constraints on the annual percentage change in TAC. These plans have been discussed, evaluated and adopted on a stock-by-stock basis, involving different timing, procedures, stakeholders and scientists involved, disregarding mixed-fisheries interactions (ICES WGMIXFISH, 2012). The technical basis of the individual management plans is detailed in the relevant stock section. All of these plans are no longer used as the basis of advice and to set TACs for a variety of reasons, including benchmarks that have revised perceptions and reference points and the extension of stock areas, rendering these plans outdated.

With the new CFP, the demand for mixed fisheries management plans covering all species caught in a fishery is increasing. EU multiannual management plans (EU MAPs) are available for demersal stocks in the North Sea (Regulation (EU) 2018/973), and demersal and deep-sea stocks in Western Waters (Regulation (EU) 2019/472), which cover stocks within WGNSSK. They are currently not used for shared stocks in the North Sea because Norway and the UK (since Brexit) have not agreed to the EU MAP. WGNSSK therefore provides advice on fishing opportunities according to the MSY or precautionary approach.

2.2.4 Additional technical measures

The national management measures concerning the implementation of the available quota in the fisheries differ between species and countries. The industrial fisheries are subject to regulations for the bycatches of other species (e.g. herring, whiting, haddock, cod) including maximum bycatch rates and technical measures on selective panels to reduce bycatch.

In 2019, the EU progressively banned the use of electric pulse fisheries in European waters (EU, 2019/1241). The total ban took effect in July 2021. Pulse fishing had been adopted by a total of 84 Dutch vessels targeting sole. The vessels have since returned to beam trawling, switched to other gears or have been decommissioned. The latter resulted in a substantial drop in the number of Dutch vessels operating in the North Sea.

Technical measures relevant to each stock are listed in each stock section, along with additional management measures, e.g. real-time closures or Fully Documented Fisheries (FDF).

2.2.4.1 Minimum landing size/Minimum conservation reference size

After the implementation of the landing obligation minimum landing sizes were renamed to Minimum Conservation Reference Sizes (MCRS) and applied from 2016 onwards. The current MCRS can be found in Table 2.2.4.1. Individuals below MCRS have to be landed but are not allowed to be sold for human consumption.

Table 2.2.4.1. Current MCRS. MCRS regions are defined in Regulation (EU) 2019/1241.

Species	MCRS North Sea	MCRS Skagerrak and Kattegat
Cod	35 cm	30 cm
Haddock	30 cm	27 cm
Saithe	35 cm	30 cm
Pollack	30 cm	–
Whiting	27 cm	23 cm
Sole	24 cm	24 cm
Plaice	27 cm	27 cm
Nephrops	85 mm (25 mm)	105 mm (32 mm)

2.2.4.2 Minimum mesh size

Regulations on mesh sizes are more complex than those on landing sizes, as they differ depending on gears used, target species and fishing areas. Many other accompanying measures are implemented simultaneously with mesh sizes. They include regulations on gear dimensions (e.g. number of meshes on the circumference), square-mesh panels, and netting material. The most relevant mesh size regulations are listed in EC No 2056/2001 and updated in Regulation (EU) 2019/1241.

2.2.4.3 Closed areas

Twelve-mile zone

Beam trawling is not allowed in a 12 nm wide zone along the British coast, except for vessels having an engine power not exceeding 221 kW and an overall length of 24 m maximum. In the 12-mile zone extending from the French coast at 51°N to Hirtshals, Denmark, trawling is not allowed for vessels over 8 m overall length. However, otter trawling is allowed to vessels of a maximum of 221 kW and 24 m overall length, provided that catches of plaice and sole do not exceed 5% of the total catch. Beam trawling is only allowed for vessels included in a list that has been drawn up for the purposes. The number of vessels on this list is bound to a maximum, but the vessels on it may be replaced by other ones, provided that their engine power does not exceed 221 kW and their overall length is 24 m maximum. Vessels on the list are allowed to fish within the twelve-mile zone with beam trawls having an aggregate width of 9 m maximum. To this rule, there is a further derogation for vessels having shrimping as their main occupation. Such vessels may be included in the annually revised second list and are allowed to use beam trawls exceeding 9 m total width.

Plaice box

To reduce the discarding of plaice in the nursery grounds along the continental coast of the North Sea, an area between 53°N and 57°N has been closed to fishing for trawlers with an engine power of more than 221 kw (300 hp) in the second and third quarter since 1989, and for the whole year since 1995. The maximum aggregated beam length in the plaice box is also restricted to 9 m. Beare *et al.* (2013) conducted a thorough analysis of the potential effect of the plaice box on the stock of plaice and concluded that no significant effect, neither positive nor negative, could be related to the implementation of the plaice box.

Sandeel box

In light of studies linking low sandeel availability to poor breeding success of kittiwake, ICES advised in 2000 for a closure of the sandeel fisheries in the Firth of Forth area east of Scotland. All commercial fishing was excluded, except for a maximum of 10 boat days in each of May and June for stock monitoring purposes. The closure was initially designated to last for three years but has been repeatedly extended and remains in force. The level of effort of the monitoring fishery was increased in 2006. In 2024, industrial sandeel fisheries are no longer allowed in English waters of the North Sea and all Scottish waters.

Norway pout box

The Norway pout fishery intensified in the northern North Sea during the 1960s and 1970s, and the concerns raised here about bycatch of juvenile cod, whiting, haddock, and saithe led to the establishment of the "Norway pout box" closed management area along the Scottish coast to protect juvenile gadoids in particular. In 1977, the UK government decided to establish this area of closure to the small-mesh trawl fishery along the eastern Scottish coast in the northern North Sea (Bigné *et al.*, 2019). Since then, the small-mesh trawl fishery has been completely forbidden in this area, with the declared aim of protecting juveniles of larger gadoid species (i.e. cod, haddock, and whiting).

Natura 2000

To protect habitats, several Natura 2000 areas have been defined. The EU currently has a plan on the table to prohibit all trawl fishing in these areas from 2030 onwards.

Unilateral management

In addition to the EU-wide statutory regulations, some countries impose additional management schemes on their fleets. One example of this is the Scottish Conservation Credits scheme which encompasses technical regulation and temporary spatial closures in return for derogation from some EU effort controls. This scheme and others are described in the stock sections to which they pertain.

2.3 Multispecies considerations

The analysis of biological interactions (predator-prey relationships) among species has been a central theme in ICES over the last 30 years, primarily for the Baltic Sea and the North Sea. The 2011, 2014, 2017, 2020, and 2023 North Sea key run performed by the multispecies group WGSAM represents the current state-of-the-art in terms of multispecies assessment, with the dynamic estimation of predation mortality.

The single-stock assessments and advice presented in the WGNSSK 2025 report are not produced by the multispecies assessment model, but time-varying values of natural mortalities estimated by multispecies assessments for cod, haddock and whiting in the North Sea are incorporated in the assessments of these species. Natural mortalities taking into account multispecies interactions as estimated in specific research also included in the single-stock assessment for Norway pout being similar to the multispecies assessment values. Flatfish have not been part of the multispecies assessment in the past. However, since the 2023 key run, plaice has been included as a prey species, although predation is at the moment limited to grey seals. More work is needed to incorporate further information on flatfish in the multispecies advice.

2.4 Stock status overview

The main impression in recent years is that fishing pressure has been reduced substantially for many North Sea stocks of roundfish and flatfish compared to the beginning of the century. Six stocks with agreed reference points (category 1 stocks) have SSB higher than MSY $B_{trigger}$ (haddock in 4, 6.a and 20, plaice in 4 and 20, sole in 4, turbot in 4, whiting in 4 and 7.d and witch in 3.a, 4 and 7.d). The SSB of saithe in 3.a, 4 and 6 is just above B_{lim}. Additionally, Northern Shelf cod (cod in 4, 6.a, 7.d and 20) has two substocks (Northwestern and Viking) with SSB below MSY $B_{trigger}$, and one substock (Southern) below B_{lim}. The SSB of Norway pout in 4 and 3.a, sole in 7.d and plaice in 7.d is below B_{lim}. Several North Sea stocks are exploited at or below F_{MSY} levels (haddock in 4, 6.a and 20, plaice in 4 and 20, sole in 4, sole in 7.d, turbot in 4, whiting in 4 and 7.d, and witch in 3.a, 4 and 7.d); however, plaice in 7.d, saithe in 3.a, 4 and 6, and all substocks of Northern Shelf cod (cod in 4, 6.a, 7.d and 20) are fished above F_{MSY}. An important feature is that recruitment remains poor compared to historical average levels for most gadoids, although there have been signs of strong recruitment for whiting in most recent years. However, recruitment in 2024 appears to be still low for the gadoids and lower than previous years for flatfish stocks, with the exception of plaice in 4 and 20, turbot in 4, and whiting in 4 and 7d.

WGNSSK is also responsible for the assessment of several data-limited stocks (category 2+ stocks) that are mainly bycatch in demersal fisheries; brill in 3.a, 4 and 7.d–e, lemon sole in 3.a, 4 and 7.d, dab in 3.a and 4, flounder in 3.a and 4, turbot in 3.a, whiting in 3.a, grey gurnard in 3.a, 4 and 7.d, striped red mullet in 3.a, 4, 7.d, pollack in 3.a and 4 and saithe in 7-10. Annual advice is required for brill in 3.a, 4 and 7.d–e, turbot in 3.a, and lemon sole in 3.a, 4 and 7.d. Biennial advice is required for striped red mullet in 3.a, 4 and 7.d, flounder in 3.a and 4, grey gurnard in 3.a, 4 and 7.d (due in 2026), and whiting in 3.a (due in 2026). Triennial advice is now required for dab in 3.a and 4, saithe in 7-10, and pollack in 3.a and 4 (due in 2027).

Category 2–6 fish stocks: In 2025, new advice has been produced for bll.27.3a47de and tur.27.3a (category 2), for dab.27.3a4, fle.27.3a4, lem.27.3a47d, mur.27.3a47d (category 3), and pok.27.7-10 (category 5). Advice was produced following the WKLIFE X rules. Advice was not provided for gug.27.3a47d, whg.27.3a (category 3), and pol.27.3a4 (category 5).

The summary of stock status is as follows:

Category 1 stocks

- Cod (cod.27.46a7d20): Spawning-stock biomass of the Northwestern substock has been above MSY $B_{trigger}$ in the most recent years and is just below in 2025. Spawning-stock biomass of the Viking substock has been fluctuating between MSY $B_{trigger}$ and B_{lim} in the most recent years and is below MSY $B_{trigger}$ in 2025. Spawning-stock biomass of the Southern substock has been below B_{lim} since 2017. Fishing pressure has been high throughout the time-series but decreased sharply from 2018 and is currently above F_{MSY} for all three substocks. Fishing pressure is also above F_{pa} for both the Viking and Southern substock. Recruitment (age 1) in recent years has been low for all three substocks, with 2025 being among the lowest recruitment in the time-series for all substocks.
- Haddock (had.27.46a20): Spawning-stock biomass has increased sharply since 2019 and is now well above MSY $B_{trigger}$. Fishing pressure has declined since the beginning of the 2000s, but it has been above F_{MSY} for most of the time-series. Since 2021, fishing pressure

has been below F_{MSY}. Recruitment (age 0) since 2000 has been low with occasional larger year classes. Three recent year classes (2019, 2020, and 2022) have all been above the recent mean. Recruitment in 2024 is estimated to be low.

- Norway pout (nop.27.3a4): Spawning-stock biomass has been fluctuating over the time-series with a period of higher biomass in the 1990s, late 2000s, and around 2020. In the most recent year, SSB decreased, and is in 2025 below B_{pa} and B_{lim}. No reference points for fishing pressure or for MSY $B_{trigger}$ have been defined for this stock. Recruitment was below average in 2021 and is very low in 2023, 2024 and 2025.
- Plaice (ple.27.420): The spawning-stock biomass is well above MSY $B_{trigger}$ and has markedly increased since 2008, following a substantial reduction in fishing pressure since 1999. Currently, fishing pressure on the stock is below F_{MSY}. Recruitment (age 1) in 2022 is the second highest in the time-series, while recruitment in 2024 is the third highest.
- Plaice (ple.27.7d): The spawning-stock biomass has increased rapidly from 2010 following a period of higher recruitment between 2009 and 2019. However, since 2016, SSB has gradually declined and is now at a level just below B_{lim}. Higher fishing pressure from 2015 onwards and lower recruitment from 2021 onwards contributed to this decrease in SSB. Fishing pressure on the stock is currently situated above F_{MSY}. Recruitment (age 1) 2024 appears to be low.
- Saithe (pok.27.3a46): Spawning-stock biomass has mostly decreased since the early 2000s and is currently just above B_{lim}. Fishing pressure has decreased and stabilized above F_{MSY} since 2000. Currently, fishing pressure on the stock is above F_{MSY}. Recruitment (age 3) has shown an overall decreasing trend over time but slightly increasing in 2022. Recruitment in 2024 is low.
- Sole (sol.27.4): The spawning-stock biomass has fluctuated between MSY $B_{trigger}$ and B_{lim} since 2005, but is above MSY $B_{trigger}$ since 2021. Fishing pressure has been fluctuating above F_{MSY} for a large part of the time-series but sharply decreased from the beginning of the 2020s. In 2024, fishing pressure is below F_{MSY}. Recruitment (age 0) in 2018 was the 5th highest of the time-series. In more recent years, recruitment is estimated to be lower. Recruitment in 2024 is the highest since 2018.
- Sole (sol.27.7d): The spawning-stock biomass has decreased since 2014 and is estimated to be fluctuating around B_{lim} since 2017. Currently, SSB is estimated to be below B_{lim}. Fishing pressure has decreased since 2007, stabilised around F_{MSY} from 2019 onwards and is currently estimated below F_{MSY}. Recruitment (age 1) is estimated to be on a relatively low level since 2012 and the 2024 recruitment is slightly increasing compared to the preceding two years.
- Turbot (tur.27.4): The spawning-stock biomass has increased since 2004 and has been above MSY $B_{trigger}$ since 2007. Currently, spawning stock size is above MSY $B_{trigger}$. Fishing pressure has been above F_{MSY} from 2018-2021, but decreased after to levels below F_{MSY}. Recruitment (age 1) is variable without a trend. In 2024, it is estimated to be higher than the preceding 4 years.
- Whiting (whg.27.47d): Spawning-stock biomass has increased substantially in recent years and is now well above MSY $B_{trigger}$. Fishing pressure has been below F_{MSY} since the early 1990s. Currently, fishing pressure on the stock is well below F_{MSY}. Recruitment (age 0) has been fluctuating without trend, but the 2019 and 2020 year classes are estimated to be the largest since 2002. Recruitment in 2024 is estimated above average.
- Witch (wit.27.3a47d): Spawning-stock biomass has decreased and has been below MSY $B_{trigger}$ since the end of the 1990s but is above MSY $B_{trigger}$ in 2025. Fishing pressure has been above F_{MSY} since the beginning of the time-series. Currently, fishing pressure on the stock is at F_{MSY}. Recruitment (age 1) has decreased since 2019 and is at lower levels in 2024.

Category 2 stocks

- Brill (bll.27.3a47de): Catches declined sharply in recent years. The relative exploitable biomass (B/B_{MSY}) has been above the reference point since the beginning of the time-series. Relative fishing pressure (F/F_{MSY}) has decreased since 2000 and is currently below F_{MSY}.
- Turbot (tur.27.3a): Catches peaked in the late 1970s and early 1990s and have been more stable in recent years. The relative exploitable biomass (B/B_{MSY}) has been above reference point since the beginning of the time-series. Relative fishing pressure (F/F_{MSY}) peaked in the late 1970s and has been relatively low without a trend in more recent years. Currently, fishing pressure on the stock is below F_{MSY}.

Category 3 stocks

- Dab (dab.27.3a4): The stock size indicator has decreased since 2016 but is still above $I_{trigger}$. Currently, fishing pressure on the stock is below the $F_{MSY\ proxy}$.
- Flounder (fle.27.3a4): The stock size indicator has gradually decreased since 2008 and is below $I_{trigger}$ in 2025. Currently, fishing pressure on the stock is below the $F_{MSY\ proxy}$.
- Grey gurnard (gug.27.3a47d): rollover advice.
- Lemon sole (lem.27.3a47d): The stock size indicator has decreased since 2016 and is below $I_{trigger}$ in 2024. Currently, fishing pressure on the stock is below the $F_{MSY\ proxy}$.
- Striped red mullet (mur.27.3a47d): The stock size indicator has fluctuated above $I_{trigger}$ in the most recent years and is above $I_{trigger}$ in 2024. Currently, fishing pressure on the stock is above the $F_{MSY\ proxy}$.
- Whiting (whg.27.3a): rollover advice.

Category 5 stocks

- Saithe (pok.27.7-10): Catches have decreased steadily since 2013. Catches in 2024 are the lowest of the time-series.
- Pollack (pol.27.3a4): rollover advice.

2.5 Ecosystem overviews

2.5.1 General observations

The latest version of the Greater North Sea Ecosystem Overview was published in April 2024 (ICES, 2024b). No new audit was done in 2025.

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